NESTED-SPACE COSMOLOGY

The Geometry of a Universe

Regions, boundaries and what an interior observer can learn

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Abstract

Nested-Space Cosmology asks how a region's geometry, fields and surrounding structure determine the world available to an observer inside it. We develop this question as a finite atlas: a parent with several black-hole boundaries, possible unequal children and further generations. Local spherical geometry provides tractable boundaries and observer charts; it does not make a multi-hole parent globally spherical or establish the topology of our universe. Published spacetime solutions, junction laws, branching mathematics and regional reduction supply the foundations. The work then concentrates on missing connections between them.

One connection is explicit. A fixed de Sitter/Schwarzschild wall joins a regular expanding cap to a vacuum bridge and an asymptotic end on a checked spacelike slice. A vacuum-end gluing theorem supplies a conditional multi-body initial-data family preserving that body. In the isolated spherical reference, the center, moving wall and right exterior have different causal access: selected early signals connect them, while new emissions after the chosen initial slice do not. A prescribed early conformal scalar pulse crosses the wall and produces a calculable transient at the center. Its waveform alone does not establish a unique exterior: the incident state is an independent input. We now identify a finite characteristic preparation in the retained white-hole sector and a bounded exterior-to-wall transfer operator. Its existence is certified analytically. Evaluating it predicts a central pulse followed by a small positive scattering tail within the declared finite window. An independent positive-kernel argument fixes the tail's sign; spatial and clock refinements test its numerical magnitude. For this fixed geometry, a complete compatible signed central history also determines a unique preparation. Positivity and a prescribed source cutoff exclude some histories; geometry comparison requires those input restrictions to be justified independently. The canonically normalized probe adds no singular surface stress or tangential flux jump. With fixed corner geometry and null normalization, the same preparation requires a calculable correction to the initial areal radius and its affine slope. Transverse constraint transport then determines a geometric mass response on that initial null surface, including a negative quadratic coefficient after the compact input. Coupled bulk and wall evolution remain open. A separate finite spherical field model demonstrates regional transfer, geometric feedback and reduced memory, without identifying its mode-space complement with a parent universe.

These results connect known mathematics in stated domains. They establish neither regular black-hole formation of children nor a complete cosmology. Physical inheritance of constants, quantum-state selection, persistent gravitational effects, antimatter asymmetry and an arrow of time remain open. The atlas makes those gaps explicit and identifies calculations that can resolve them without requiring every microscopic question to be solved first.

I. The question and the spherical idea

1. A universe viewed from within

An observer measures clocks, fields, curvature and correlations within a limited domain. Such measurements need not uniquely identify the larger structure containing that domain. Our question is whether this ambiguity can become a constructive research programme: assemble possible surroundings, determine which attachments satisfy their equations, and ask what each permits an interior observer to experience.

The organizing sequence is **borrow established mathematics, connect compatible regions, investigate branching and inheritance, calculate local consequences, then compare with observations**. A universal action tying together all fields and regions remains an ambitious candidate. It is not a prerequisite for learning whether a particular boundary can be joined or whether a signal can reach a particular observer.

A mature atlas would let a reader follow a possible universe from its surroundings, through its boundaries, into an observer's world and onward through compatible transitions. It would compare families of histories with explicit geometry, dynamics and observables. Completion means dependable coverage of declared model families, including exclusions and observational degeneracies. Establishing

that a family describes our universe is a further empirical achievement. The present finite connections are steps toward that destination, not a claim to have reached it.

The first motif contains two spatially separated black holes in a common parent, possible unequal children, and one further generation. A separate motif lets one hole have multiple offspring. These pictures suggest questions; their arrows do not supply an equation of motion, a formation mechanism or a communication channel. Figure 1 distinguishes the broad motif from the specific attachment whose mathematics has been checked.

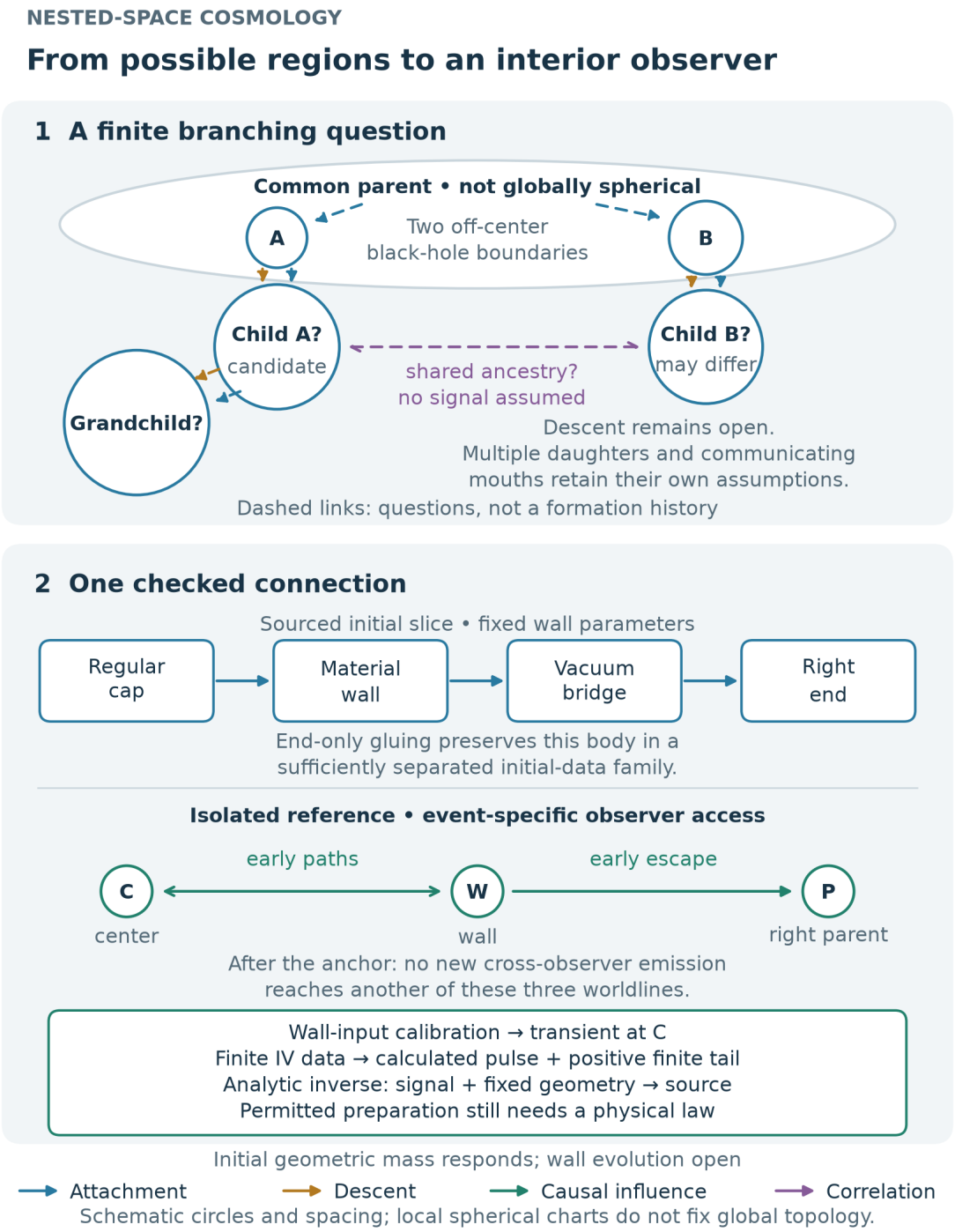


Figure 1. Possible descent (upper panel) remains hypothetical; the lower panel shows the checked cap–wall–bridge body and observer/field reference. Colors distinguish attachment, descent, influence and correlation. Circles and spacing do not establish an embedding, relative scales or global spherical topology. The figure record identifies each connection's evidence and status.

2. What spherical geometry means here

There is a useful borrowed energy argument with a precise domain: in Euclidean space, a ball minimizes boundary area at fixed enclosed volume. A positive constant isotropic surface tension therefore favors a spherical boundary when surface energy is the quantity being minimized. Direction-dependent tension instead selects a Wulff shape; see Neumayer, 1503.06705v2, Section 1.1. This motivates testing rotational symmetry. It does not prove a cosmological equilibrium, dynamical relaxation or a probability ranking over all geometries. Gravity, rotation, matter stresses and the actual boundary law must be specified before importing that variational conclusion into a spacetime model.

A spherical boundary is a two-dimensional surface with area $4\pi R^2$. A spatial interior may be a three-ball; a closed spatial universe may instead have three-sphere topology. These are different assertions. Drawing one circle inside another establishes none of them.

In a locally spherically symmetric spacetime, a useful form is

$$ds^2 = g_{ab}(x)\,dx^a dx^b + R(x)^2 d\Omega^2.$$

Here g_{ab} is a Lorentzian metric on the two-dimensional orbit space and R is areal radius. We use signature $(-+++)$ for the atlas. Radial proper distance, areal radius and a diagram's vertical coordinate need not coincide. In particular, a radial profile is not automatically a surface that can be embedded in Euclidean three-space.

The physical spherical hypothesis is that useful cosmological regions or interfaces may admit this structure, exactly or as a controlled approximation. It is not a theorem inferred from statistical isotropy of observations. Several off-center holes generally break global spherical symmetry. We therefore use local spherical charts where justified and an explicitly nonspherical parent where required. Deviations from symmetry are a question for each model, rather than a reason to discard the organizing idea.

3. One thread through three bodies of work

The continuing question is: **how do a region's fields, boundary and surroundings shape its state, geometry and measurements?**

Body of work	What it contributes	Connection still missing
Preserved foundation	Spectral inheritance proposal, boundary operators, source and response constructions	A physical inheritance law and common cosmological source completion
Finite spherical companion	A finite realization of regional transfer, geometric feedback and memory with declared action and state	Identification with an actual assembled parent–child spacetime
Atlas evidence and this living manuscript	Geometric compatibility, a sourced slice, observer access, a calibration response and an evaluated finite exterior preparation	Remote preparation, later evolution, other attachments and observational discrimination

Their mathematics is complementary. A finite operator's eliminated modes are not automatically the exterior spacetime beyond a black hole. The atlas must first identify a field equation, state, clock and matching law before an operator reduction can acquire that geometric meaning.

The foundation's useful historical calculations remain in their source-bound edition and owning notes. This manuscript changes the organizing story: geometry and observer access lead; stronger unifying claims follow only when their missing connections are supplied. No priority claim is made for junction conditions, Schur complements, branching processes or the published wall solutions used below.

II. Regions, inheritance and the meaning of a connection

4. Four arrows that must not be confused

Each region in the atlas carries a geometry, source or state, clock convention, dimensionless parameters and validity domain. Each connection states the borrowed mathematics and the additional condition still needed.

Connection	Mathematical question	What it does not establish by itself
Descent	Where do a child's initial data originate?	A traversable path
Geometric attachment	Which boundaries share admissible metric and source data?	Formation or stability
Causal influence	Can an intervention reach an observer within the stated domain?	A nonzero field response without state and coupling
Correlation	Which information is shared through state or ancestry?	Communication between siblings

Two regions may inherit correlated data without ever exchanging a later signal. An attached interior can lie beyond the causal reach of a specified parent observer. Conversely, communicating mouths require their own causal and material assumptions; they cannot be inserted into a disconnected offspring model as an unlabeled extra arrow.

The multiple-daughter geometry of Barrabès and Frolov (hep-th/9511136v1), the charged semiclassical multi-mouth route of Emparan et al. (2012.07821v2), and quantum-state coupling distributions discussed by Giddings and Turiaci (2004.02900v2) remain separate library entries. A state's coupling distribution is not itself a signal exchanged between its branches. Their source cards and unresolved connections are recorded in the mathematical library.

5. A precise finite inheritance construction

Evidence: exact finite record, frozen snapshot, and proof and domain.

The foundation proposes an inherited normalized law, $T_\Theta^* D_\Theta = D_\Theta$. One controlled meaning is available without postulating an infinite cosmos. Fix a Hermitian d-dimensional matrix H, a nonzero link B, a positive scale ratio Ω, and a finite depth N. For regional labels $j=0,\dots,N-1$, define a block Hermitian matrix by the following entries where their indices exist:

$$(J_{m,N})_{jj} = \Omega^{m+j} H$$
$$(J_{m,N})_{j,j+1} = \Omega^{m+j} B.$$

The lower off-diagonal blocks are the adjoints. Let U_m consistently relabel the equal-sized regional frames. Then

$$J_{m+1,N} = \Omega U_m J_{m,N} U_m^\dagger$$

$$\Omega^{-1} U_m^\dagger J_{m+1,N} U_m = J_{m,N}.$$

Each block acquires the same extra factor, so the identity follows directly. For fixed depth,

$$\operatorname{spec}(\Omega^{-m} J_{m,N}) = \operatorname{spec}(J_{0,N}).$$

This is a finite realization of normalized inheritance under chosen scaling data. It does not derive Ω, establish a physical transition through a horizon, or show that the universe selects this law. Changing N changes the boundary problem and can change the dressed spectrum. The finite identity is not a proof of cosmic natural selection.

The shared law also does not fix identical states. A fermionic covariance $0 \le C \le I$ evolves as

$$C(t) = e^{-iJt} C(0) e^{iJt}.$$

Regional occupations and cross correlations can differ under the same J. Thus “same laws,” “same state,” and “same observed constants” require separate arguments. A physical transition would have to determine the relevant state and dimensionless data, rather than obtaining them from the normalization identity alone.

6. What an unresolved surrounding region can leave locally

Split a specified operator into retained and complementary blocks A, D with link B. For an admissible complex spectral parameter z, elimination gives the usual self-energy

$$\Sigma(z) = B(zI - D)^{-1} B^{\dagger}.$$

The retained resolvent is $[zI - A - \Sigma(z)]^{-1}$. Nested elimination is associative when the required inverses exist. For the finite inherited chain its normalized recurrence is

$$\Gamma_j(x) = xI - H - \Omega^{-1} B \Gamma_{j+1}(x/\Omega)^{-1} B^{\dagger}.$$

The factor $1/\Omega$ is essential. This is borrowed finite linear algebra applied to the declared family, not an additional force law. To interpret Σ as a cosmological exterior response requires a physical operator, field state and boundary map. Sections 14–15 give a separate finite Hilbert-space realization of this reduction, without identifying its complementary modes with a cosmological exterior.

III. A finite map of possible geometries

7. Start with two holes, without manufacturing two children

The explicit parent benchmark uses time-symmetric Brill–Lindquist vacuum initial data,

$$h_{ij} = \psi^4 \delta_{ij}$$
$$K_{ij} = 0$$
$$\psi = 1 + \frac{m_A}{2r_A} + \frac{m_B}{2r_B}.$$

The recorded choice is $m_A = 1$, $m_B = 1.5$, separation $d = 12$. It has outer ADM mass 2.5 and two inner-end masses 1.0625 and 1.5625. These are three asymptotically Euclidean ends of an initial slice. Extra ends do not by themselves become cosmological children. The companion hole supplies nonspherical structure, so an exact isolated Schwarzschild collar cannot simply be assumed at either puncture. The parent record and owning analysis retain the construction and checks.

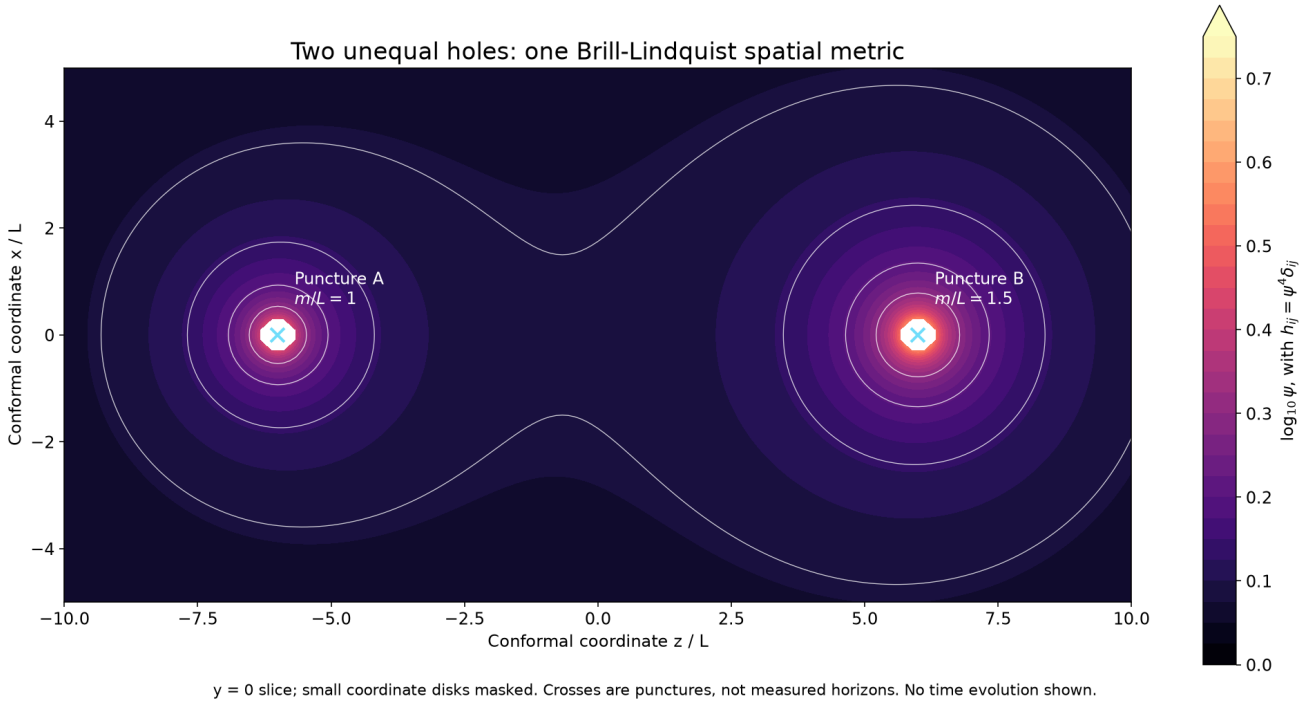


Figure 2. The explicit two-hole initial-data benchmark. This is a nonspherical common parent with unequal ends, not a solved evolution that creates children. Local spherical models must earn an attachment to it or belong to a separate parent family.

The alternative parent route imports an existence result instead of re-solving its gluing equations. Chruściel, Corvino and Isenberg (0909.1101v1) provide sufficiently separated scalar-flat bodies with retained interior regions and a common Schwarzschild outer end. Their more general vacuum-end theorem (1004.1355v1) allows appropriate non-time-symmetric ends. These establish separate families, not a modification that leaves the displayed Brill–Lindquist data unchanged. No explicit sufficient separation is certified here.

8. A failed attachment can sharpen the map

Evidence: radiation junction, moving-boundary conservation/control, and source assumptions.

The first attachment screen pairs a closed radiation FLRW region with a vacuum exterior and a pure constant-tension wall. For a comoving boundary, $a=A\sin\eta$, $t=A(1-\cos\eta)$, and $R=a\sin\chi$. The junction condition would require, over an interval of changing R,

$$k^2R^4 - 2\cos\chi\, kR^3 + 2MR - A^2\sin^4\chi = 0.$$

For nonzero $A\sin\chi$, this cannot be a polynomial identity on that interval. The recorded $\chi=\pi/3$, $M=0.4A$ test exposes a source incompatibility, not a failure of all interiors.

Could a moving wall repair the same model? Intrinsic conservation answers before a new simulation is needed. Fixed pure tension has zero intrinsic stress divergence, so the Codazzi relation requires zero net normal energy flux. In the fluid rest frame,

$$T(U,n) = (\rho + p)\gamma^2 v.$$

A vacuum exterior contributes no flux and radiation has $\rho+p>0$; therefore $v=0$. Motion alone cannot rescue these assumptions. This is a compatibility application of an established conservation law. Allowing a different surface source, exterior state or flux would define a different candidate requiring its own test.

9. A distinct fixed-source wall that does connect

For the next branch, take de Sitter inside and Schwarzschild outside,

$$f_- = 1 - H^2R^2$$
$$f_+ = 1 - \frac{2M}{R}.$$

In $G=c=1$ units let $\kappa=4\pi\sigma>0$, with surface pressure $P=-\sigma$. The signed angular extrinsic curvatures obey

$$\beta_- - \beta_+ = \kappa R$$
$$\beta_\pm = \varepsilon_\pm \sqrt{\dot{R}^2 + f_\pm}\,.$$

Here the dot is wall proper time and $\varepsilon_\pm$ fixes the normal orientation. Squaring and then forgetting those signs would lose physical branches. The useful explicit expressions are

$$\beta_- = \frac{M}{\kappa R^2} + \frac{\kappa^2 - H^2}{2\kappa} R$$

$$\beta_+ = \frac{M}{\kappa R^2} - \frac{\kappa^2 + H^2}{2\kappa} R$$
$$\dot{R}^2 = \beta_-^2 - f_-\,.$$

We hold $H\ell=1$, $\kappa\ell=2$, $M/\ell=0.4$ fixed. On the tested expanding segment $1.2\le R/\ell\le1.8$, $\beta_->0$ and $\beta_+<0$. The wall proper duration is 0.3718706103ℓ. Direct unsquared and pressure checks, including a separate Christoffel-based calculation, agree within the recorded numerical residuals. The wall's areal-radius rate is not a local three-speed. This published wall family is classified more fully in Section 11; the source has not been retuned to make the segment work.

10. A sourced initial slice and a conditional common parent

Evidence: slice constraints, parent assembly, slice derivation, and end-only gluing argument.

At $R_w=1.5\ell$, the interior is a flat expanding de Sitter cap: a regular spatial three-ball with $h=dl^2+R^2d\Omega^2$, $R'=1$ and $K^i_{\ j}=-H\delta^i_{\ j}$. We use $K=-L_T\,h/2$, where T is the future unit slice normal. Define $V=T(R)$. The cap has $V_-=HR$. Its slice determines the wall boost,

$$\gamma = \frac{\beta_- - HR\dot{R}}{1 - H^2R^2}$$
$$\gamma v = \frac{\dot{R} - HR\beta_-}{1 - H^2R^2}.$$

The exterior inherits

$$R'_+ = \gamma\beta_+ - \gamma v\dot{R}$$
$$V_+ = \gamma\dot{R} - \gamma v\beta_+\,.$$

At this event $\gamma=1.00926203$, $v=0.13516599$, $R'_+=-2.02778609$, and $V_+=1.90925372$. The exterior slice is therefore not time symmetric at the material wall.

A fixed smooth slicing function tapers $V(R)$ to zero at $R_0=\ell$, before the Schwarzschild throat $2M=0.8\ell$. In the transition,

$$R' = -\sqrt{f_+ + V^2}$$
$$K^{\theta}{}_{\theta} = -\frac{V}{R}$$
$$K^{l}{}_{l} = -\frac{dV}{dR}.$$

These data describe a slice of the same vacuum Schwarzschild spacetime. They do not interpolate the matter source. Its mass identity is

$$M = \frac{R}{2}(1 + V^2 - R'^2).$$

Where $V=K=0$, the slice continues smoothly through a bridge using $R=2M\cosh^2 u$, with the cap on the negative-u side and the selected parent end on the positive-u side.

The distributional shell constraints are equally necessary. The surface energy and radial momentum on this slice are $E_\Sigma=\sigma\gamma$, $J_\Sigma=\sigma\gamma\, v$, giving

$$[R'] = -4\pi R E_\Sigma$$
$$[V] = 4\pi R J_\Sigma.$$

Their recorded residuals are below 2×10^{-15} in normalized units; independent finite-difference bulk residuals are below 10^{-10}. These are implementation checks, not a global stability estimate.

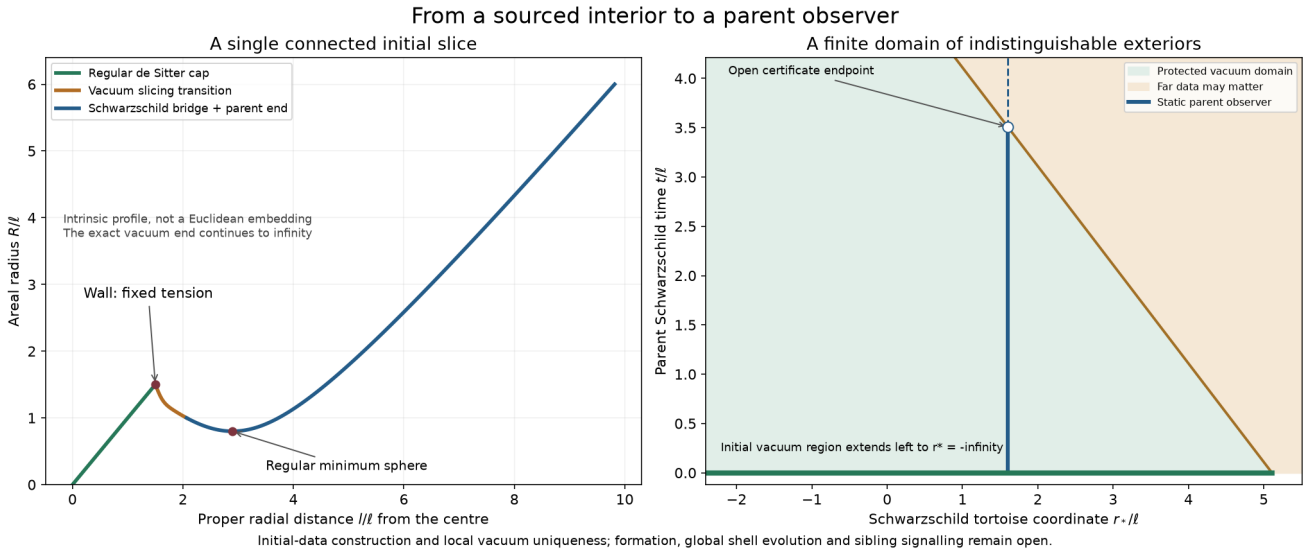


Figure 3. The verified initial connection. Spatial distances are not ages. Because $|R'|>1$ on part of the chosen slice, the radius profile should not be interpreted as a Euclidean surface of revolution. The protected right-exterior observer interval is a separate causal-domain statement.

The selected right end of this body A, of mass 0.4ℓ, can be used with the selected right end of an ordinary complete Schwarzschild bridge B, of mass 0.6ℓ, in the imported vacuum-end gluing theorem. It acts on smooth vacuum ends and preserves an open overlap. Reattaching unchanged interiors on identical full (h,K) data leaves the original shell delta source intact and creates no seam source. We do not apply a smooth global-data corollary directly to a thin shell.

The resulting conditional initial-data family has one designated common parent end **and the untouched other end of** *B*: two asymptotic ends in total. That second end has not become a cosmological child. Sufficiently large separation is required but no numerical threshold or explicit multi-body development has been calculated. The achievement is a compatible sourced initial connection, not a regular formation history, a general Einstein–shell evolution theorem or a stable cosmological future.

IV. What the observers can reach

11. Complete one reference before assigning its causal arrows

Evidence: isolated causal completion, observer access, and protected parent interval.

The wall's isolated spherical completion is the flat-exterior limit of solution 13 in Aguirre and Johnson, gr-qc/0508093v2, Figures 7 and 10. Writing $x=R/\ell$, its expanding branch has

$$\dot{R}^2 = \frac{25x^2}{16} - 1 + \frac{3}{10x} + \frac{1}{25x^4} > 0.$$

It starts at a finite proper-past $R=0$ boundary in the Schwarzschild white-hole sector, crosses $R=2M$ into the left exterior, and crosses the de Sitter horizon into an expanding interior chart. It reaches arbitrarily large radius at infinite wall proper time. Its Schwarzschild curvature $48M^2/R^6$ diverges at the past boundary. The regular cap on the chosen slice therefore does not imply a regular past formation. The retained reference also contains the usual future Schwarzschild black-hole singular boundary.

Choose three clocks. C is the cap center, with de Sitter flat proper time t_C. W carries wall proper time τ_W, set to zero at $R=1.5\ell$. P is static in the right exterior at $R_P=1.6\ell$, with proper time $\tau_P=\sqrt{1-2M/R_P}\ t_{Schw}$. Their origins coincide with a selected slice convention; this is not an equality of universe ages.

For the flat de Sitter chart,

$$\eta = -\ell e^{-t_C/\ell}$$
$$R = ar$$
$$a = e^{t_C/\ell}.$$

Radial null lines have constant $\eta-r$ or $\eta+r$. At W_0, $\eta=-\ell$ and $r=1.5\ell$. Its inward ray would reach $\eta=0.5\ell$, outside the chart's future $\eta<0$; hence it never reaches C. A central signal emitted at $t_C=0$ likewise cannot catch the future wall.

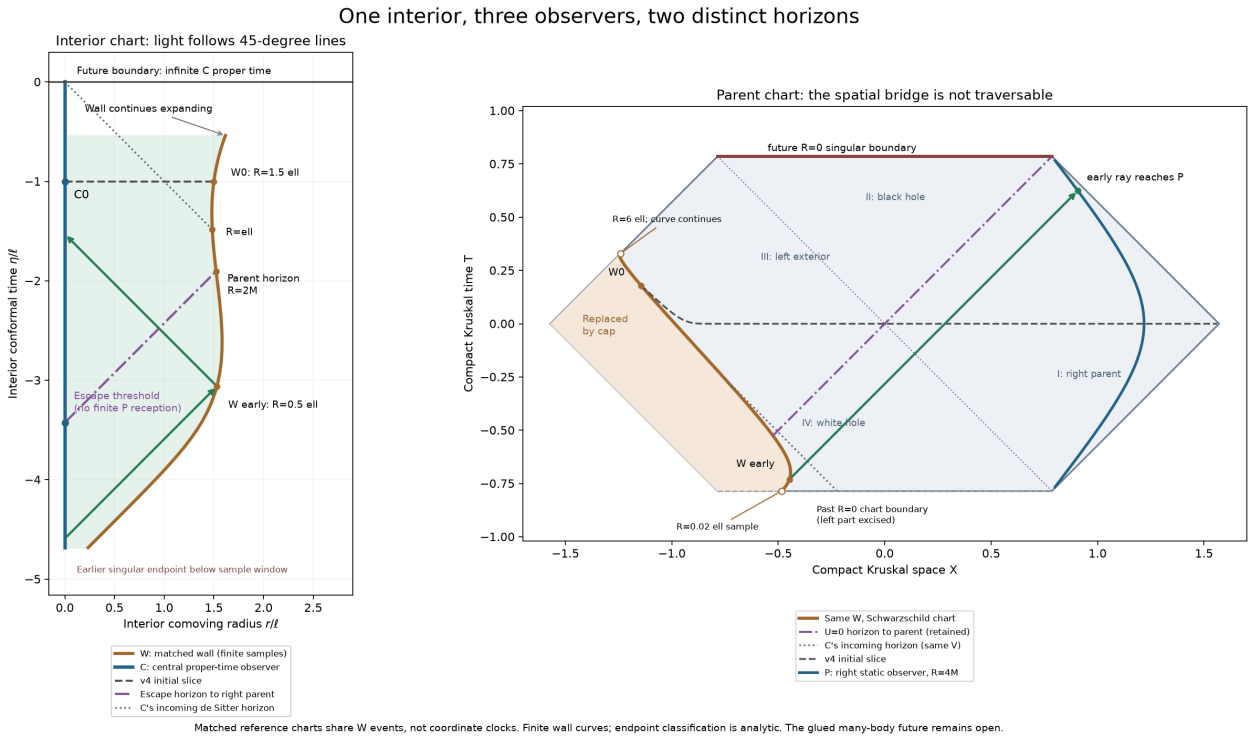


Figure 4. The isolated reference in joined causal charts. Horizons are clipped to retained domains; the central cosmological horizon continues into the Schwarzschild chart along its appropriate constant null coordinate. The faint past R=0 line also shows uncut chart context, part of which is excised. Open wall endpoints at R/ℓ=0.02 and 6 mark finite plot samples, not physical beginnings or endings.

Emission and destination	Result in this reference	Required distinction
New $C\to W$ or $W\to C$ emission at or after the anchor slice	No future reception	Earlier signals may arrive later
Right parent $P\to W$ or $P\to C$	No future causal route	Left and right exterior regions are different
Early $W\to C$, while $R<\ell$	Allowed	Reception time depends on the emission event
Early $C\to W\to P$, crossing while $R<2M$	Allowed	The limiting horizon ray has no finite parent reception
These same arrows in a glued multi-body future	Unresolved	Initial-data gluing does not supply that global future

For a wall event with interior time t_W, the central emission reaching it and the reception of its inward ray satisfy

$$t_e = t_W - \ell \ln(1 + R/\ell)$$
$$t_r = t_W - \ell \ln(1 - R/\ell).$$

The second relation requires $R<\ell$. At the early event $R=0.5\ell$, the recorded central emission is -1.5242168241ℓ and inward reception is -0.4256045355ℓ. An outward early route through that event reaches P at its own proper time $+2.5311656510\ell$. Thus exclusion of **new emissions** after a slice must never be rewritten as exclusion of all later receptions. Emissions approaching the central horizon from below can arrive arbitrarily late.

The critical central emission for escape to the right parent is $t_C=-1.2312944525\ell$. Equality follows the parent horizon and has no finite parent reception. That horizon extends into the cap along η-$r=\eta_W(2M)$-$r_W(2M)$; it is not the surface $R=2M$ everywhere. The central observer's de Sitter horizon

$\eta+r=0$ is a different boundary. Detailed clock domains and finite checks are retained in the observer note.

12. Different exteriors can be locally indistinguishable

There is also a result that does not require the glued parent's full future. Retain the exact right Schwarzschild $K=0$ slice for $2M<R<R_{cut}$, with $R_{cut}=10M$. For a static observer at $R_{obs}=4M$, the exterior cutoff cannot influence it within

$$0 < t_{\mathrm{Schw}} < r_*(R_{\mathrm{cut}}) - r_*(R_{\mathrm{obs}})$$
$$r_*(R) = R + 2M\ln(R/(2M) - 1).$$

The observer's proper duration is the right side multiplied by $\sqrt{1-2M/R_{obs}}$. For $M=0.4\ell$, the bound is $t_{Schw}<3.5090354889\ell$, or $\tau_P<2.4812627896\ell$. The inner optical boundary is at negative infinity, so it introduces no finite inner cutoff.

Shared smooth vacuum data give the same vacuum development in this domain. The owning argument uses standard Cauchy-development uniqueness, with Sbierski (1309.7591v2) as a source. The bound ends a guaranteed interval; it does not predict that a signal actually arrives then. Matching additional fields and their states requires additional data.

This supplies one precise atlas of indistinguishable exteriors. Distinct larger assemblies can share a protected local geometry. The next useful question is which later measurement or extra state datum separates them. Failure to distinguish exteriors can be a result, rather than a reason to add unconstrained parameters until a preferred story appears.

V. Fields, transfer and finite memory

13. A first field that reaches the interior

Evidence: field transfer and checks, distributional matching, and finite initial-state construction.

A causal path permits a response; a field law and state determine one. On the fixed wall geometry, take a classical real massless scalar in four dimensions, its spherical $l=0$ sector, and conformal coupling $\xi=1/6$. Its action and equation are

$$S_\phi = -\tfrac{1}{2}\int \sqrt{-g}\,[(\nabla\phi)^2 + \xi\mathcal{R}\phi^2]\,d^4x$$
$$(\Box - \xi\mathcal{R})\phi = 0.$$

There is no independent scalar interaction added to the wall. With the normal directed from cap to exterior, the pure-tension jump gives $[K]=-3\kappa$ and singular curvature $6\kappa\delta$. Distributional matching therefore requires

$$[\phi] = 0$$
$$[n\phi] = 6\xi\kappa\phi.$$

For $\chi=R\phi$, using $[nR]=-\kappa R$,

$$[\chi] = 0$$
$$[n\chi] = (6\xi - 1)\kappa\chi.$$

Conformal coupling cancels this radial delta barrier. It does not remove the smooth Schwarzschild scattering potential $f_+ 2M/R^3$. The jump law is borrowed from the distributional field equation; see Santiago et al., 1512.02120v2, equations 31–33. Their static charged-shell setup is not the moving solution used here.

Inside the conformally flat cap, regularity at the center gives

$$\chi = F(\eta + r) - F(\eta - r)$$
$$\phi_C = \frac{2F'(\eta)}{a}.$$

The outgoing term enforces central regularity; it is not a second independently adjustable central source. The conformal reduction follows the published field transformation of Huguet, Queva and Renaud (gr-qc/0603031v1).

Set the initial interior field and its normal derivative to zero on the cap slice where the wall has $R=0.05\ell$. Prescribe one exterior-side incoming characteristic at the wall,

$$I = (d/d\tau_W + n)\chi.$$

The plus derivative extracts the inward wave; its propagation direction is U_W-n. No global exterior state or quantum vacuum is specified. For the calibration pulse let $\delta=0.03\ell$, $\tau_*=-1.1138558046\ell$, and $z=(\tau_W-\tau_*)/\delta$. Define $B(z)=(1-z^2)^4$ for $|z|<1$, with $B=0$ outside, and set

$$I = \frac{\varepsilon}{\delta} B'(z).$$

The extended B is C^3, and I is C^2, not an infinitely smooth bump. The amplitude ε is a formal test-field normalization. The incoming trace is stipulated zero outside the pulse through the retained response interval.

Let $T(\tau_W)$ be central arrival proper time. On this early branch,

$$T = t_W - \ell \ln(1 - R/\ell)$$
$$T' = \frac{\gamma(1 + v_{\text{pec}})}{1 - R/\ell} > 0.$$

The exact conditional central transfer is

$$\phi_C[T(\tau_W)] = \frac{I(\tau_W)}{T'(\tau_W)}.$$

This relation includes the wall boost and reception-clock Jacobian. It is not a geometric-optics approximation.

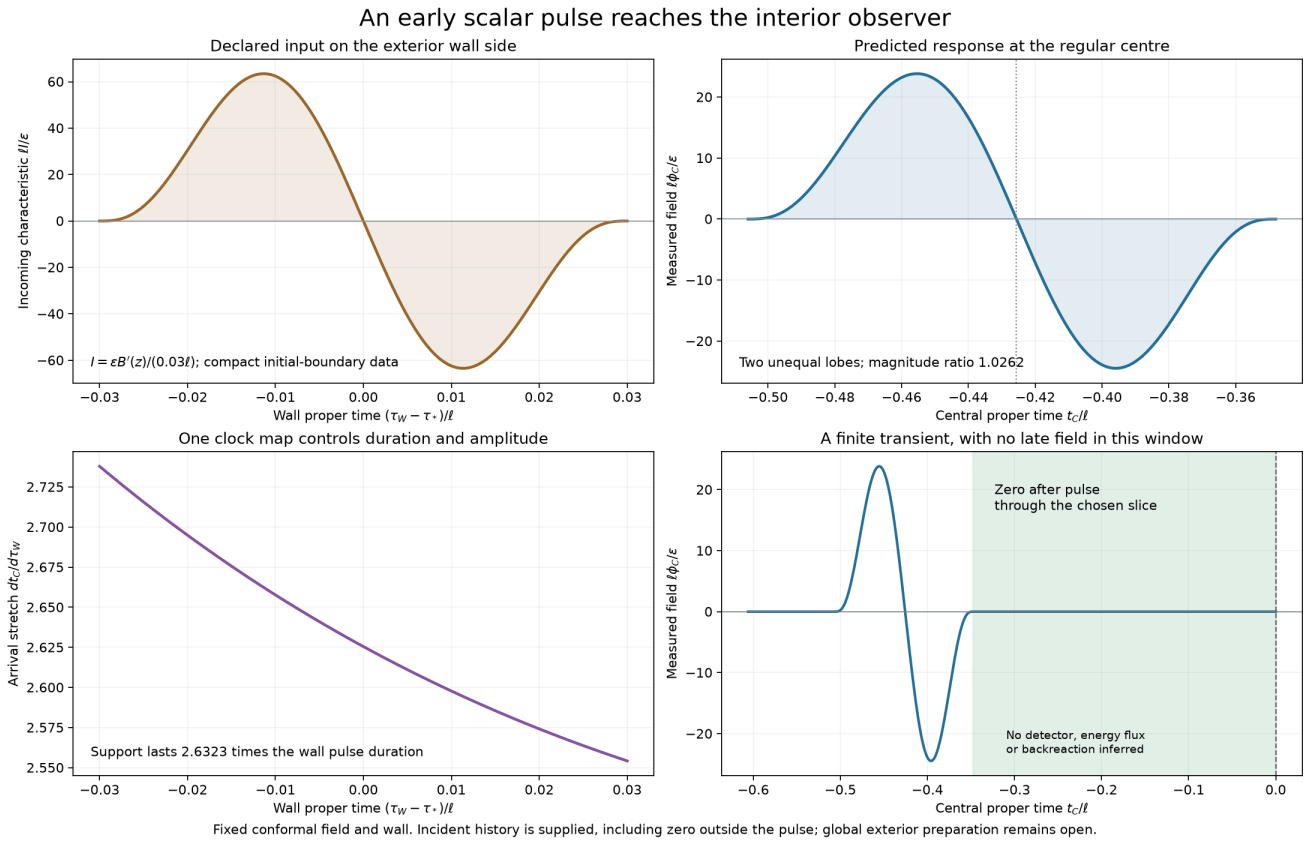


Figure 5. A prescribed classical pulse, propagated on the fixed geometry. The source record normalizes input as ℓ I/ε and response as ℓφ$_C$/ε. The incoming history is a calibration state. Its remote preparation, energy flux, quantum state and backreaction have not been calculated.

The wall support lies at *R/ℓ=0.47499005* to *0.52285042*. The central response lies at t_C/ℓ*=-0.50593397* to *-0.34799584*, giving a duration stretch of *2.63230219*. Its normalized peaks are *+23.84144800* and *-24.46533478*; the magnitude ratio is *1.02616816*. These are waveform amplitudes, not particle abundances or energy-transfer efficiencies.

The center field is exactly zero after this received support through t_C*=0*, conditional on the stipulated incoming history. From the first central arrival until that endpoint, the wall's comoving radius is at least *0.8ℓ*, while the earliest reflected ray reaches at most *0.65853382ℓ*. The *0.14146618ℓ* gap excludes a cap round trip in this interval. It does not exclude an uncomputed exterior tail under different incoming data, erase an observer's record, or imply that the whole field vanishes everywhere.

There is explicit freedom in this wall-input calibration. On an admissible monotone arrival interval, a desired central trace *Y(T)* can be produced by *I(τ)=T'(τ)Y[T(τ)]*, subject to the same regularity and state restrictions. Within the zero-area pulse family, the corresponding trace must also have zero area. This inverse fixes a wall input under the stated transfer law; it does not construct two compatible exteriors with the same observation. The right parent observer cannot prepare it through a causal channel in this reference. The following construction supplies independent finite data elsewhere in the retained geometry; their remote physical origin remains unspecified.

Minimal coupling remains a distinct alternative: *[nχ]=-κχ* and the cap radial equation contains *-2χ/η*^{2}. Its known bulk de Sitter tail does not solve the moving-boundary problem. The conformal transient therefore neither proves nor excludes memory in other field theories.

13.1. Where the early signal can originate

Evidence: finite preparation and bounds, derivation and independent checks.

Keep the same geometry and conformal probe. The right-exterior observer *P* still cannot signal the wall or center. However, the early wall is adjacent to retained Schwarzschild white-hole geometry. Finite data

there can reach it without introducing a source at the singular past boundary. This is an exterior-side preparation, not a signal sent by P.

In dimensionless Kruskal coordinates $U,V<0$, future radial causal curves have nondecreasing coordinates. With $x=R/\ell$, define

$$F(x) = (1 - x/0.8)e^{x/0.8} = UV.$$

Choose wall endpoints $x_a=0.45$, $x_b=0.55$ and radial floor $x_f=0.4$. Set $V_a=V_W(x_a)$, $V_b=V_W(x_b)$ and $U_0=F(x_f)/V_a$. The retained preparation domain is

$$D = \{V_a \le V \le V_b, \quad U_0 \le U \le U_W(V)\}.$$

The wall truncates this domain: a rectangular continuation across it would use the wrong field equation. Monotonicity of F gives $0.4\le R/\ell\le 0.55$ throughout D, away from both $R=0$ and $R=2M$. Prescribe zero cap field and time derivative at the earlier flat time $t_a=t_W(x_a)$, and zero reduced field on the bottom characteristic $V=V_a$. On $U=U_0$, independently prescribe $g(V)=\varepsilon\, B(4(V-V_m)/(V_b-V_a))$, where $V_m=(V_a+V_b)/2$ and B is the C^3 bump above. It vanishes near the corners. Neither the outgoing wall characteristic nor a desired received waveform is independently specified.

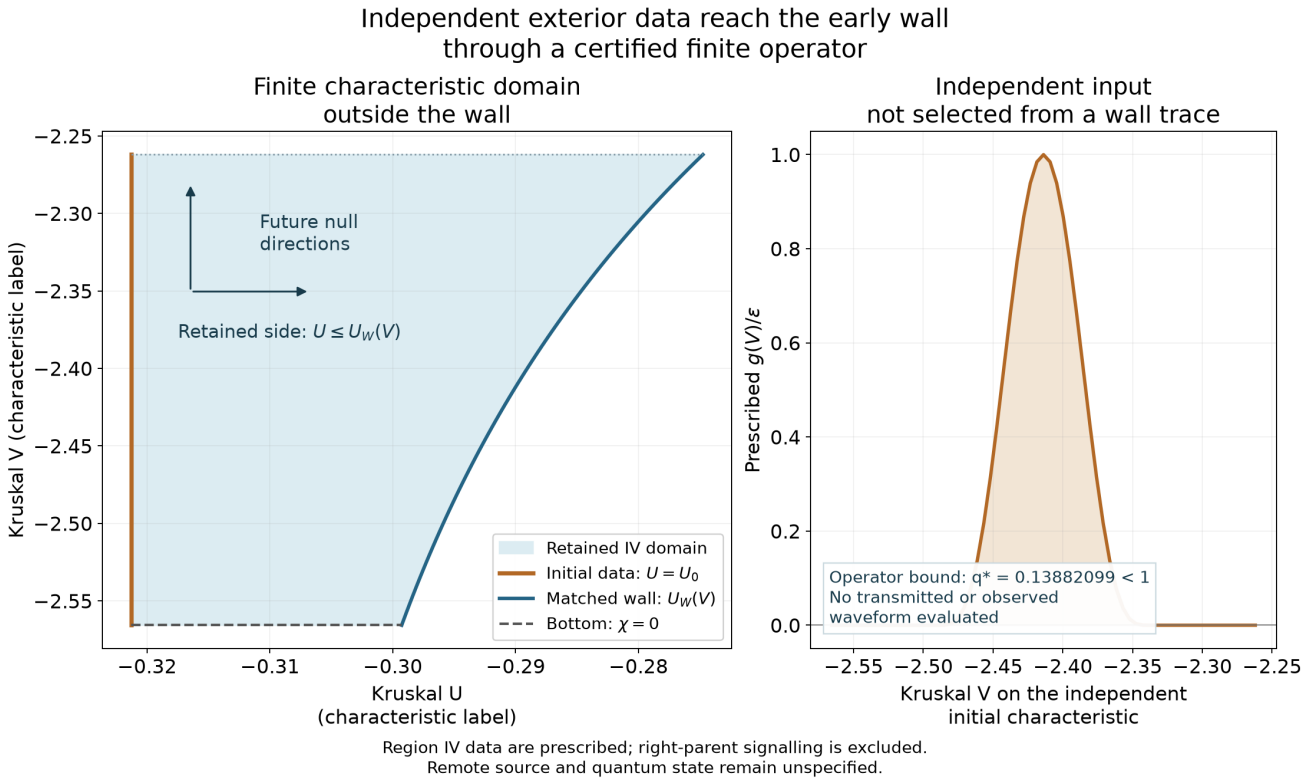


Figure 6. The preserved v7 preparation checkpoint, before the waveform evaluation in Section 13.2. The finite preparation occupies the retained side of the early wall in sector IV. The profile on the characteristic surface is input, not a calculated wall or central waveform. The singularity and horizons lie outside this finite domain. Figure 4 supplies the global chart context.

The cap null labels $u=\eta-r$ and $v=\eta+r$ increase along the timelike wall. The earliest possible reflected signal has outgoing label v_a, whereas $u_b<v_a$ throughout this wall window; the recorded gap is $(v_a-u_b)/\ell=2.63459191$. No central reflection can yet return. The normal toward the retained exterior has components

$$n^U = -\dot{U}_W$$
$$n^V = \dot{V}_W.$$

The vanishing outgoing characteristic consequently supplies $\chi_U=0$ on the wall.

The borrowed Schwarzschild scalar potential is f_+2M/R^3; see Dolan and Ottewill, 1106.4318v1, Eq. 11. Its usual asymptotic scattering conditions are not imported into sector IV. The local operator becomes

$$\chi_{UV} + K\chi = 0$$
$$K = \frac{16M^4}{R^4} e^{-R/(2M)} = -\frac{R_{UV}}{R}.$$

Let $L(u)=V_a$ for $u\leq U_a=U_W(x_a)$ and otherwise let $L(u)$ be the inverse wall graph. The complete reduced problem is

$$\chi(U,V) = g(V) - \int_{U_0}^{U} du \int_{L(u)}^{V} dv\, K(u,v)\chi(u,v).$$

Differentiating recovers the PDE, characteristic data and wall condition. An analytic bound on the integral operator in the supremum norm is

$$q_* = 16e^{-1/2}\left[\sqrt{F(0.4)} - \sqrt{F(0.55)}\right]^2 \simeq 0.138821 < 1.$$

The bound uses $K\leq 16e^{-1/2}$ and an AM–GM bound on the enclosing area, independent of numerical clock quadrature. The standard contraction principle therefore gives a unique finite solution. Moreover,

$$\|\chi - g\|_\infty \leq \frac{q_*}{1-q_*}\|g\|_\infty.$$

At the positive input peak the wall reduced field is at least 0.8388ε. This bound is not a central-field peak or an energy-transmission efficiency.

The wall input is now determined by the solution:

$$I = 2\dot{V}_W\left[g'(V_W) - \int_{U_0}^{U_W(V_W)} K(u,V_W)\chi(u,V_W)\,du\right].$$

The existing relation $\varphi_C[T(\tau_W)]=I/T'$ then defines the observer response over $-0.581759\leq T/\ell\leq -0.250231$. This closes the finite data-to-wall-to-observer operator. Section 13.2 evaluates it without imposing the calibration's late zero field. Remote generation of g, quantum-state selection, backreaction and regular formation of the wall remain distinct open connections.

13.2. A prepared pulse leaves a finite scattering tail

Evidence: evaluated preparation and response, derivation, admission and controls.

The next calculation keeps the same geometry, source profile, wall law and linear conformal probe. It determines the signal from independent data, rather than choosing a wall trace to reproduce a desired central waveform. Write the positive integral operator in Section 13.1 as A. Reversing its integration order gives

$$(A\chi)(U,V)=\int_{V_a}^{V}dq\int_{U_0}^{\min\{U,U_W(q)\}}dp\,K(p,q)\chi(p,q).$$

This form exposes a causal evolution in V. For numerical bookkeeping only, extend χ constantly in U beyond the moving wall. All kernel integrals remain clipped at the wall; this does not apply the Schwarzschild equation inside the cap. The directly evaluated wall derivative is

$$J(V)=g'(V)-\int_{U_0}^{U_W(V)}K(U,V)\chi(U,V)\,dU$$

$$\frac{\ell\phi_C}{\varepsilon}=\frac{2J/\varepsilon}{d(T/\ell)/dV}.$$

We integrate χ-g with borrowed DOP853 time stepping and a clipped trapezoidal spatial integral. The wall crossings of grid nodes delimit steps, preserving the partial-cell boundary integral. No new field solver family or asymptotic scattering condition is introduced. The owner records the initial stopped attempt and this event-aligned integration correction.

The sign of the resulting tail has an independent analytic check. For positive input g and kernel K, reversing the nested integrals gives $0\le A^2g\le q_*Ag$ pointwise. Positivity then bounds subsequent terms in the alternating Neumann series. After the input has ended,

$$-Ag\le\chi\le-(1-q_*)Ag<0\quad(U>U_0).$$

Consequently $J=-\int K\chi\,dU>0$: scattering in the retained exterior leaves a positive central field during this post-input window. Defining $B(V)=\int K(U,V)(Ag)(U,V)\,dU$ supplies a quantitative bracket,

$$(1-q_*)B(V)\le J(V)\le B(V).$$

The transverse kernel integral is available exactly. With $m=M/\ell=0.4$ and x_a,x_b the radii at $U=a,b$ on the same V,

$$\int_a^b K(U,V)\,dU=\frac{2m^2}{V}\left(x_b^{-2}-x_a^{-2}\right).$$

Independent quadrature of the remaining two integrals checks the bracket at three fixed post-input times. It does not reuse the evolved field.

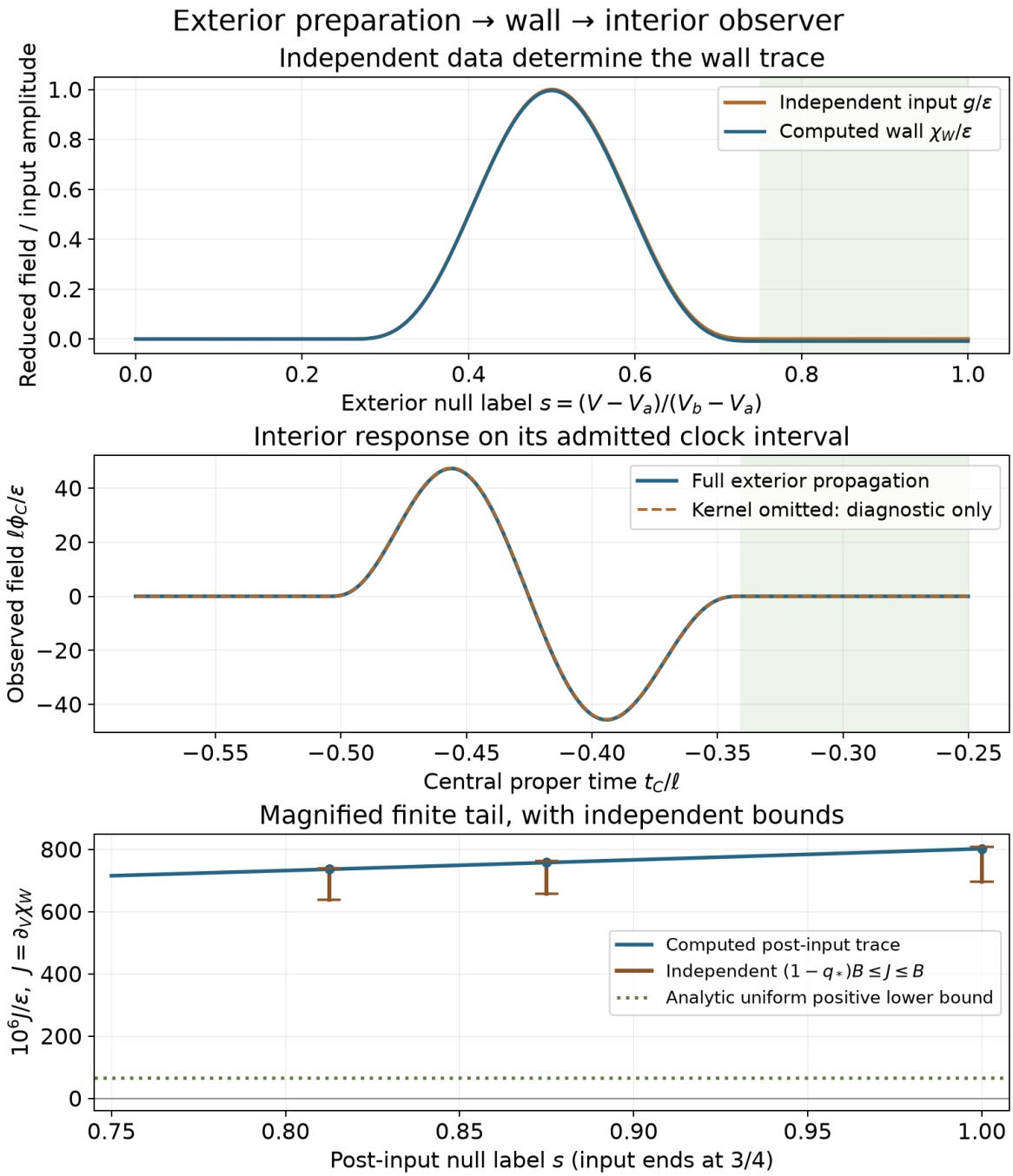


Figure 7. Independent characteristic data produce a wall trace and central response. The lower panel resolves the small positive post-input derivative with independent analytic brackets. The zero-potential curve is an integration diagnostic on the same coordinate map, not a second physical spacetime. The plotted tail stops at the admitted domain boundary.

Finite result or check	Value
Sampled positive / negative central extrema, $\ell\varphi_C/\varepsilon$	*47.29834 / -45.69596*
Central field at the final admitted arrival	*0.001290498*
Maximum central change, 257 to 513 spatial nodes	2.79×10^{-7}
Maximum central change, 129 to 257 clock knots	3.41×10^{-8}
Final J/ε and independent bracket	*0.000802039* in *[0.000696588,0.000808877]*

These sampled extrema are field amplitudes, not energy efficiencies or certified continuum extrema. The positive post-input interval is $-0.340773<T/\ell\le-0.250231$. Spatial refinement, a separate clock refinement and a constant-kernel Bessel benchmark support the numerical evaluation. The recorded experiment uses 5,597,264 kernel evaluations, within its ten-million evaluation and 60-CPU-second limits. Numerical comparisons are diagnostics; the sign conclusion rests on the positive-kernel argument, not a claimed rigorous floating-point enclosure.

This closes one missing connection: finite exterior data now yield a checked observer response, including an exterior-generated tail even though the conformal cap has no bulk tail. It does not establish a lasting remnant, backreaction, a signal from P, or an observation identifying a unique exterior. The useful next question is which response features survive freedom in the preparation. Remote generation and continuation beyond this window remain separate open problems.

13.3. What the signal determines when preparation is unknown

Evidence: analytic inverse record, proof and preparation restrictions.

The forward response now permits a sharper inference question. Hold the geometry, scalar law, zero cap and no-return window fixed. Write $f=\chi/\varepsilon$, $g=f(U_0,V)$, $h=f(U_W(V),V)$, $\theta=T/\ell$ and $Y=\ell\varphi_C/\varepsilon$. These are signed classical fields, rather than particle counts or intensity-only measurements. The inherited central map can be inverted by integration:

$$h(V)=\tfrac{1}{2}\int_{\theta_a}^{\theta(V)} Y(s)\,ds.$$

The zero initial cap fixes its integration constant. To recover the exterior preparation, subtract the forward wedge equation at the wall from its value at U. With E extending a function of V constantly in U, this yields

$$f=Eh+\mathrm{B}f$$
$$(\mathrm{B}f)(U,V)=\int_U^{U_W(V)}dp\int_{L(p)}^{V}dq\,K(p,q)f(p,q).$$

Every integration point stays in the retained exterior. The existing operator-norm bound is $q_*<1$, so the standard resolvent supplies

$$f=(I-\mathrm{B})^{-1}Eh$$
$$g(V)=f(U_0,V).$$

Direct differentiation recovers $f_{UV}=-Kf$ and the zero outgoing wall characteristic; restriction recovers the source, prescribed wall trace and zero bottom. The inverse and forward constructions are mutual inverses. The change in the derivative of L at $U_a=U_W(V_a)$ requires a corner check, supplied explicitly in the owning proof.

More precisely, take C^2 source traces that vanish on the initial interval $[V_a,V_s]$, where $V_s=V_a+(V_b-V_a)/4$. They correspond bijectively to C^1 central histories vanishing on $[\theta_a,\theta(V_s)]$. The initial zero interval removes corner mismatches. This enlarged class allows signed inputs with arbitrary later shape; it does not require the original middle-half support, positivity or polynomial profile. The result uses the exact smooth geometry and known strictly increasing proper-time map. It concerns the reduced exterior field and regular-centre trace; higher regularity of the full scalar at the centre requires stronger input regularity. For two histories in this same model,

$$\|g_1 - g_2\|_\infty \le \frac{\theta_b - \theta_a}{2(1-q_*)}\|Y_1 - Y_2\|_\infty .$$

This controls source amplitude under errors in complete compatible field histories. It supplies no bound on source energy, derivatives, backreaction, or uncertainty in the geometry. Numerical reconstruction from sampled data has not been performed.

Physical restrictions on the preparation change the allowed response range. For any nonzero nonnegative source supported in the original $[V_s,V_e]$, the positive-kernel argument of Section 13.2 implies, for $V_e \le V \le V_b$,

$$h(V) < 0$$
$$Y(\theta(V)) > 0$$
$$\int_{\theta_a}^{\theta(V)} Y(s)\,ds = 2h(V) < 0.$$

A smooth, nonzero, nonnegative observer pulse supported strictly inside $(\theta(V_s),\theta(V_e))$ violates this condition. Its integrated h instead has a positive plateau; positivity of the inverse resolvent forces $g \ge h > 0$ after V_e. Such a trace has a preparation in the enlarged class, but that preparation cannot obey the original cutoff. This is an exact exclusion within the fixed model, rather than another numerical waveform.

The result therefore separates reconstructing a source from identifying geometry. Two geometries are observationally degenerate only when a trace lies in both physically allowed response ranges. No second geometry is constructed here. Controlled nonlinear source-response maps can carry much richer geometric information, even with one receiver, as illustrated by Tzou, 2102.01841v3; those assumptions are absent from this passive linear experiment. Comparing candidate exteriors needs an independent preparation restriction or a source-linked additional channel. The next section tests a prerequisite for a physical preparation: whether its stress is compatible with the existing wall.

13.4. The transmitted probe and the wall's stress

Evidence: analytic stress record, derivation and independent junction check.

The normalized waveform leaves its absolute gravitational stress unspecified. Fix the scalar kinetic coefficient to one, zero scalar potential and no direct scalar-wall interaction. In the same Jordan frame the curvature coefficient is

$$f(\phi) = \frac{1}{2\kappa_4} - \frac{\xi\phi^2}{2}$$
$$\kappa_4 = 8\pi G, \quad \xi = \tfrac{1}{6}, \quad \phi = \varepsilon\Phi.$$

Background vacuum stress remains independent of the scalar. The material tension, geometry and original C^3 compact preparation remain fixed on the finite domain of Section 13.2. This regularity supplies the one-sided derivatives needed here; the broader inverse class alone does not establish full four-dimensional stress regularity at the center. Assume $f>0$ for the weak-field interpretation, without inferring a backreaction bound.

The borrowed nonminimal action gives the full improved Hilbert stress:

$$T^{\mathrm{can}}_{\mu\nu} = \nabla_\mu\phi\nabla_\nu\phi - \tfrac{1}{2}g_{\mu\nu}(\nabla\phi)^2$$
$$T^{\phi}_{\mu\nu} = T^{\mathrm{can}}_{\mu\nu} + \xi\left[G_{\mu\nu}\phi^2 + (g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu)\phi^2\right].$$

The Einstein-tensor term is counted once. With the same common normal, write $d=[n\varphi]=k\varphi$ and $C_{ab}=[K_{ab}]-\gamma_{ab}[K]=2k\gamma_{ab}$. The distributional curvature and second derivatives then give the singular scalar surface coefficient

$$S^{\phi}_{ab} = \xi\{2\phi d\gamma_{ab} - \phi^2 C_{ab}\} = 2\xi\phi(d - k\phi)\gamma_{ab} = 0.$$

An independent substitution into the coupled variational junction law of Aviles, Maeda and Martinez, 1910.07534v2, Eqs. (3.43)-(3.46) recovers the unchanged material tensor $t_{ab}=-\sigma\gamma_{ab}$. Continuous scalar coefficients make the delta terms well defined; products with discontinuous normal derivatives are avoided.

The regular tangential flux must also be checked. Using one-sided traces, constant k, the original curvature jump and the vanishing mixed Einstein components of both vacuum backgrounds gives

$$[T^{\phi}_{na}] = (1 - 6\xi)k\phi D_a\phi = 0.$$

The canonical kinetic term alone would give a nonzero jump; the improvement terms cancel it. Equality of the two fluxes does not mean either flux vanishes. Thus these two local conditions require no additional scalar surface source or tangential exchange from the material wall at quadratic order.

The smooth bulk stress remains an order-ε^2 gravitational source. No bulk metric response, normal-force balance, quantitative amplitude bound or stability result follows from the surface cancellation. Before evolving that source, its gravitational initial data must satisfy the constraints. The next section supplies one such connection; it does not yet select a physical preparation law for observer inference.

13.5. The prepared pulse requires gravitational initial data

Evidence: initialization record, constraint derivation and independent frame check.

The field was propagated on a fixed geometry. Adding its gravitational source while keeping all initial metric data unchanged would generally violate an Einstein constraint. We retain the same compact characteristic preparation and apply the spherical null constraint, borrowed from An and Lim, 2005.04090v1, Eqs. (2.6)-(2.7), with the full conformal stress of Section 13.4 replacing their matter source.

Write the smooth exterior metric as

$$ds^2 = -A(U,V)\,dU\,dV + R^2 d\Omega^2$$
$$G_{VV} = -\tfrac{2}{R}\left(R_{VV} - \tfrac{A_V}{A}R_V\right).$$

At the source-leg corner keep the reference radius $R=0.4\ell$, both expansions and the lapse. The zero-data bottom leg retains its reference geometry. On $U=U_0$, choose $A=A_0$, so that the background radius $x=R_0/\ell$ is an affine parameter with fixed corner origin and scale. This is a declared gauge and gravitational-data choice. Fixing both the reference lapse and radius everywhere on the sourced leg is not available.

Express the original compact source as $\chi=\varepsilon\, b(x)$ and put $p=b/x$. The leading scalar trace and formal radius correction are

$$\phi = \frac{\varepsilon}{\ell} p + O(\varepsilon^3)$$
$$\alpha = \frac{\kappa_4 \varepsilon^2}{\ell^2}$$
$$\frac{R}{\ell} = x + \alpha h(x) + O(\alpha^2).$$

The affine constraint fixes the coefficient rather than choosing it to fit an observation:

$$h'' = -\frac{x}{3} p'^2 + \frac{x}{6} p p''$$
$$h(0.4) = h'(0.4) = 0.$$

Primes mean *d/dx*. After the compact trace ends, integration by parts gives

$$h'(x) = -\frac{1}{2} \int_{0.4}^{x} s\, p'(s)^2 \, ds < 0$$

$$h(x) = -\frac{1}{2} \int_{0.4}^{x} (x - s) s\, p'(s)^2 \, ds - \frac{1}{6} \int_{0.4}^{x} p(s)^2 \, ds < 0.$$

Thus the finite source leaves a reduced initial areal radius and affine radial slope relative to the corner-anchored reference. The conformal stress need not be pointwise positive during the pulse for these signs to hold. The slope is not itself the null expansion $2R_{,x}/R$.

At the last source point $x_b = 0.4815042083$, the recorded coefficients are $h = -6.1228335482$ and $h' = -162.1805536249$. The calculation uses 129 displayed samples and 15,360 scalar integrand evaluations, with no new wave or bulk Einstein solve. Independent direct integration and integration by parts agree; separate quadrature and geometric refinements change h by less than 4.7×10^{-14} and h' by less than 1.7×10^{-12}. These are numerical diagnostics, not rigorous decimal enclosures or bounds on the omitted perturbative terms. No finite amplitude has been selected: the formal radius displacement is $\alpha \ell h$.

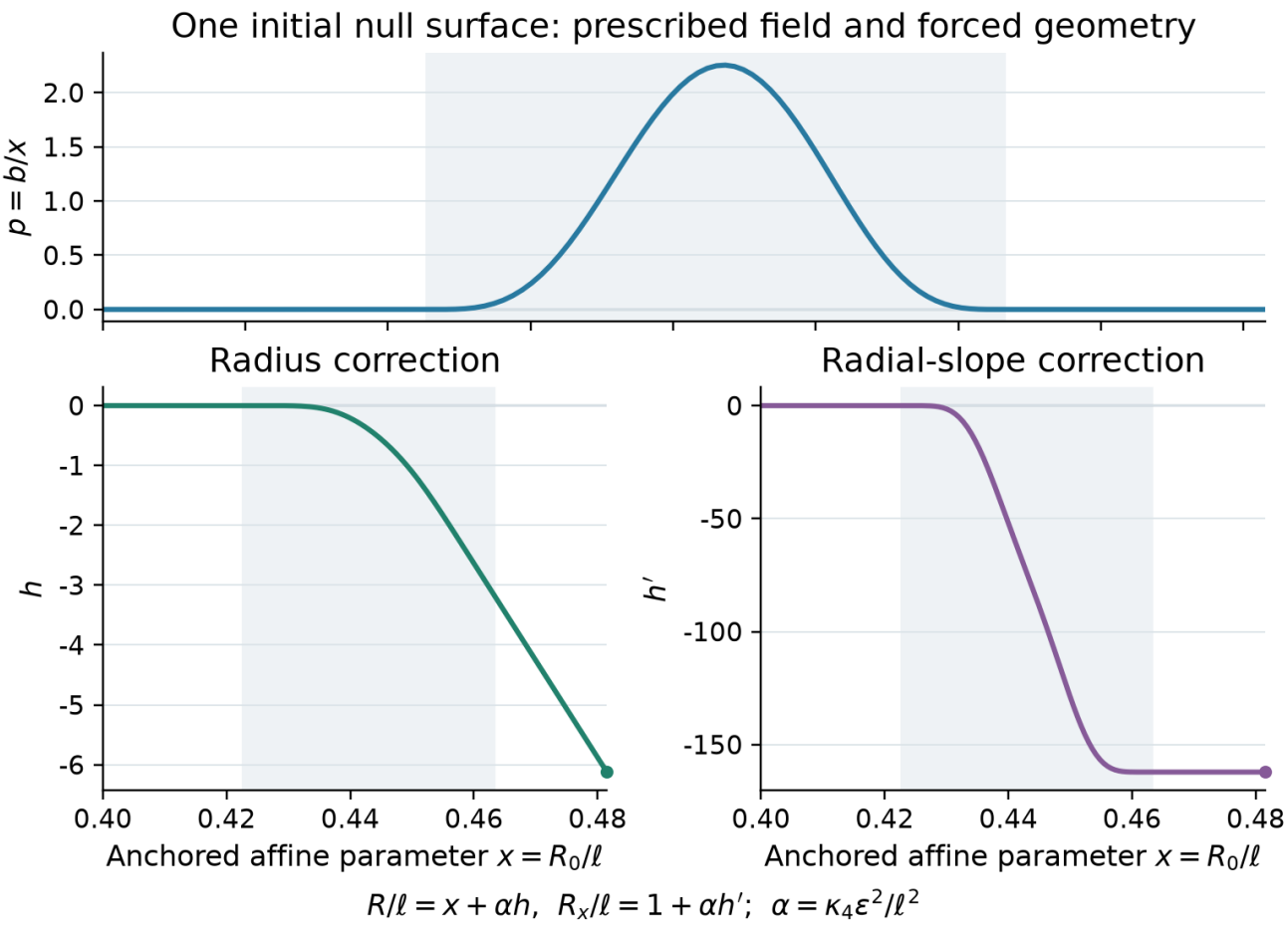


Figure 8. The original compact trace and its quadratic gravitational coefficients along one initial null leg. The affine parameter is the reference areal radius divided by ℓ, with its origin and scale fixed at the corner. Shading marks the compact input interval. After it ends, the radial-slope correction stays negative and constant. These coefficients describe initial data; they are not a mass shift or an evolved wall response. The figure record binds the plotted samples to the initialization evidence.

An independent smooth-exterior check rescales the metric by $1-\kappa_4\varphi^2/6$ and transforms the affine parameter as well. The equivalent Einstein-frame coefficient then satisfies $h_E''=-xp'^2/2$, the minimally coupled affine focusing equation. This frame comparison does not extend through the material wall without transforming its action too.

This result supplies one intrinsic characteristic constraint. By itself it does not fix the transverse geometry, later wall matching or the coupled interior response. Although $T_{VV}=0$ after the trace on this leg, transverse scalar derivatives can persist: it is not a claim that the remaining spacetime is vacuum. No mass shift, wall displacement, finite-amplitude error bound or stability conclusion follows from this intrinsic constraint alone. The next section supplies its transverse-radius connection without starting a new coupled solver.

13.6. An invariant mass response on the same initial surface

Evidence: source-leg mass record, derivation and independent stress check.

An intrinsic radius correction does not determine a spherical gravitational mass: the transverse radius derivative is also required. Its corner value was already fixed. Borrowing the mixed Einstein equation transports that datum along the same initial null surface and closes this missing connection. The quantity compared is the geometric Misner--Sharp mass length,

$$\mathfrak{m} = \frac{R}{2}\left(1 - g^{ab}\partial_a R \partial_b R\right)$$
$$\frac{\mathfrak{m}_J}{\ell} = m + \alpha\mu_J + O(\alpha^2), \quad m = 0.4.$$

It is invariant under coordinate changes. The constant reference mass also makes this quadratic coefficient unchanged by an order-α displacement of the compared cuts. A conformal change of metric generally changes its value. Dividing the length by G gives mass units when $c=1$; this does not turn a finite source-leg value into an asymptotic energy or the mass of the entire parent.

On the source leg write $ds^2/\ell^2 = -C\,dU\,dx + r^2 d\Omega^2$, where $C = -4a/U_0 > 0$, $a = 2m = 0.8$ and $r = R/\ell$. The reference transverse derivative is $d = r_{,U}/C = (a/x - 1)/4$. The leading scalar derivative $s = \ell\varphi_{,U}/(C\varepsilon)$ is supplied by the original field equation and zero bottom data:

$$s' = -\frac{s + dp'}{x}$$
$$s(0.4) = 0.$$

No additional outgoing field profile is chosen. In the equivalent smooth Einstein-frame description, the borrowed minimal mass law of An and Lim, Eq. (2.15) gives

$$\mu_E(x) = -\frac{1}{4}\int_{0.4}^{x} t(a - t)p'(t)^2\,dt.$$

Both future radial derivatives are positive in this retained white-hole sector. Consequently this coefficient decreases for a positive canonical null stress; untrapped-exterior monotonicity rules have different signs.

The return to the declared Jordan-frame geometry uses the full geometric transformation of Faraoni and Vitagliano, Eq. (7):

$$\bar{\mathfrak{m}} = \omega\mathfrak{m} - \frac{R^3}{2\omega}(\nabla\omega)^2 - R^2\nabla\omega\cdot\nabla R$$
$$\omega = \sqrt{1 - \kappa_4\phi^2/6}\,.$$

Keeping both null derivatives yields the quadratic coefficient

$$\mu_J = \mu_E + \frac{mp^2}{12} + \frac{x^2 p}{3}(s + dp').$$

The frame masses differ during the input. Independent substitution into the mixed-radius equation and the geometric mass definition reproduces this expression. A second check uses Hayward's general-matter law, Eq. (2.9d), retaining the improved mixed stress instead of setting it identically to zero as in the minimal-scalar comparison.

After the compact trace ends, $p = p' = 0$ and both derivatives of ω vanish. The frame masses then agree at this order:

$$\mu_J = \mu_E = -\frac{1}{4}\int_{\text{support}} t(a - t)p'(t)^2\,dt < 0.$$

For the recorded input, this final coefficient is approximately *-28.8831970541*. During the input the largest sampled frame difference is *4.2654598348*; this is a sampled comparison, not an extremum bound. Ten exact symbolic identities and the independent mixed-geometry and improved-stress constructions check the borrowed mass laws. The 129-sample calculation uses 47520 counted quadrature nodes; separate quadrature and geometry refinement comparisons change the mass coefficients by less than 4.6×10^{-13}. These are numerical diagnostics rather than rigorous enclosures or bounds on omitted perturbative orders.

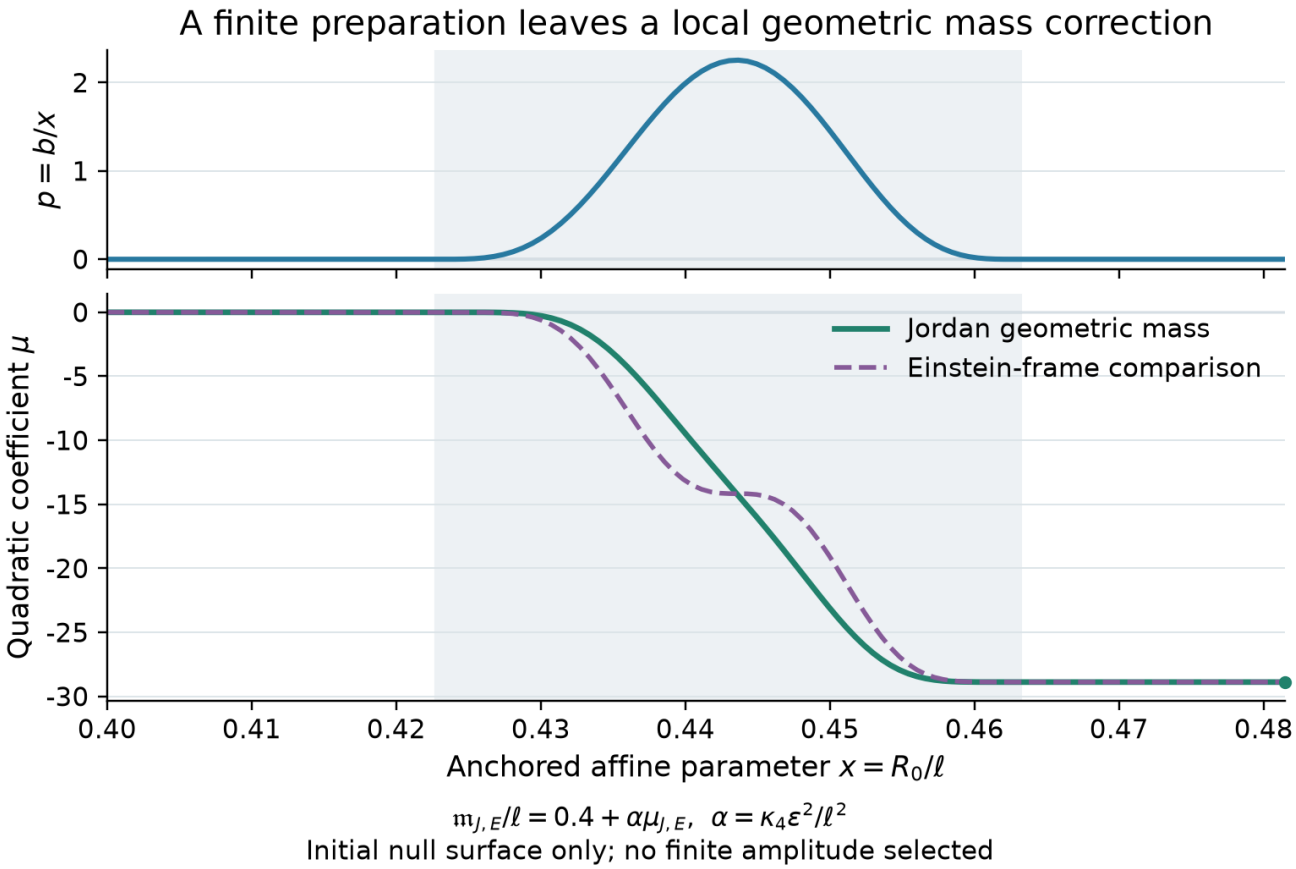


Figure 9. The fixed compact preparation (top) and its quadratic geometric mass coefficients on the same initial null leg (bottom). The horizontal coordinate retains the corner-anchored affine normalization $x=R_0/\ell$. Jordan and Einstein frame masses differ during the input and agree at a constant negative correction afterwards. The negative coefficient is relative to the reference mass; no finite amplitude or negative total mass is established. The figure record binds the plotted samples to the mass evidence.

The sign follows from *0<t<a* and the nonzero original input. The coefficient stays constant on the remaining finite source leg. Transverse scalar derivatives can still persist there, so this is neither a vacuum future nor a global mass-loss history. It is an invariant response to the specified initial data, with the frame and comparison fixed. No finite amplitude or perturbative remainder bound has been chosen. The next connection is to transport compatible sourced geometry toward the wall and test its dynamical junction conditions. Wall motion and the corrected interior-observer signal remain open.

14. A separate finite model with geometric feedback

Evidence: focused companion snapshot, finite episode, local response, and memory controls.

The finite companion supplies another part of the programme: fields transfer between selected regions while the same declared action evolves the geometry. It uses a periodic radial chart, positive chart variables and antiperiodic spinors. In the present signature convention its metric is

$$ds^2 = r^2[-L^2 dt^2 + Q^2(dx + \beta\, dt)^2 + d\Omega^2].$$

The physical lapse is rL and radial scale is rQ. Conformal gauge has $L=Q$, $\beta=0$; the reduced field Hamiltonian is $H=\sigma_2 P+\kappa\ Q\sigma_1$. Here $\kappa=1$ labels the retained angular block and is independent of the wall-tension parameter used in the atlas. Six prepared Gaussian fermion columns determine

$$C = \Phi\, \mathrm{diag}(0.75, 0.75, 0.5, 0.5, 0.25, 0.25)\Phi^\dagger$$
$$\Phi^\dagger \Phi = I_6.$$

This is a finite admissible state with $0 \le C \le I$ and $Tr\ C=3$. Unoccupied retained bands are part of the preparation, not a derived physical vacuum. The action coefficients, magnetic sector, clock and cutoff are recorded inputs.

Four matched discretizations, using 256/512 field modes and time steps 0.001/0.0005, reach coordinate $T=0.3$ without resetting the state. The reported transfer and curvature comparisons have sub-percent refinement indicators in their stated observables. The finite evolution supplies a same-action example, not a continuum error theorem, an eternal engine or the creation of new universes. Collars of this periodic geometry are regions of one evolving model.

Its local reduction keeps the original first two mode columns and eliminates their Hilbert-space complement. That complement includes modes with support inside and outside the spatial observation window. Consequently it is not the parent end of Section 10, and its Hamiltonian cannot be transferred to the wall model without a field-theoretic identification.

15. Memory is a consequence of elimination, not an added reservoir

For a time-dependent Hermitian block Hamiltonian with retained block $A(t)$, coupling $B(t)$ and complementary block $D(t)$, let $U_E(t,s)$ propagate D. Eliminating the complementary amplitude gives

$$y(t) = U_E(t, t_*)y_* - i\int_{t_*}^{t} U_E(t, s)B^\dagger(s)x(s)\, ds$$

$$\dot{x}(t) = -iA(t)x(t) - iB(t)U_E(t, t_*)y_* - \int_{t_*}^{t} K(t, s)x(s)\, ds$$

$$K(t, s) = B(t)U_E(t, s)B^\dagger(s).$$

The memory kernel, actual initial complementary drive and initial cross correlations all matter. If R_A,R_E denote the corresponding solution maps, the retained covariance is

$$C_A = R_A C_{AA} R_A^\dagger + R_E C_{EE} R_E^\dagger + R_A C_{AE} R_E^\dagger + R_E C_{EA} R_A^\dagger.$$

The companion uses the actual evolved state at $t_*=0.2$, then compares the reduced and full histories through $t=0.3$. The selected occupations change from approximately $(0.10557448,0.10557464)$ to $(0.03214496,0.03226900)$. The maximum reduced/full occupation discrepancy is 6.35×10^{-6}, compared with a physical-model change of about 0.07343.

Conditional omission control	Maximum occupation effect	Its separate numerical reference discrepancy
Omit the memory term	0.0168185	2.20×10^{-6}
Omit the actual initial complementary drive	0.0485412	1.66×10^{-5}
Omit initial cross correlations	0.0920614	1.23×10^{-5}

These controls hold the generated geometry fixed. They are not separately coupled physical evolutions; in particular, deleting the memory term is a diagnostic alteration, not a new Hermitian theory. They demonstrate why an interior description must retain the appropriate history and state information. The next bridge to the atlas would derive such a regional Hamiltonian and state from a particular matched spacetime and field, rather than labeling these finite modes as universes.

VI. What the map can ask next

16. Constants, spectral scales and physical inheritance

Open physical target. A shared normalized operator can preserve dimensionless spectral relationships while regional clocks and absolute frequency units differ. To turn this into a statement about constants, the attachment must specify clocks, rods, fields and the transition of their dimensionless parameters. Useful comparisons include α, mass ratios, and $Gm^2/(\hbar\, c)$. A numerical change caused solely by unit rescaling is not a changing law.

Successive inheritance maps also need compatible output and input spaces: the first transition must supply data on which the next law is defined. Drawing another nested sphere does not establish this composition.

The atlas wall parameters $H\ell$, M/ℓ and $\kappa\ell$ are declared inputs, not values inherited from a parent calculation. Physical inheritance might preserve a law while changing states, select among admissible states, or alter an effective environment. These possibilities need different maps and should remain distinguishable.

The foundation also contains a scalar-link construction whose invariant sector has the ordinary Dirac relation $E^2=|p|^2+\Phi^2$. It is a useful benchmark relating a coupling to a gap. It does not derive the observed particle species, nuclear masses or their interactions. Particles are not assumed to be little universes. A particle interpretation requires the appropriate field representation, observables and dynamics, not resemblance between a spectral scale and a cosmological radius.

17. Wavelength, preparation and the role of light speed

The derived scales

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}$$
$$t_P = \frac{\ell_P}{c}$$

are defined combinations of constants, as tabulated by NIST CODATA. Dimensional construction alone does not establish a smallest physical wavelength or an indivisible spatial cell. A claimed minimum would need a model of probes, observables and gravitational backreaction. Minimal-length scenarios themselves distinguish limits on resolution, distance products and other quantities; see Hossenfelder, 1203.6191v2, Sections 5.3–6.

Absolute zero refers to thermodynamic temperature and the limiting thermal state, not to an endpoint of allowed wavelengths; see NIST's temperature definition. A relation between a regional temperature and a spectral scale requires a specified state and Hamiltonian. Changing a temperature unit supplies neither.

A horizon is not automatically a reflecting cavity wall or a prohibition on longer-wavelength modes. A particle-like exterior description likewise requires an additional quantum state and measurement model; exterior mass, spin and charge alone do not establish that description. Quantum observables can have continuous as well as discrete spectra. A detector event or a wave crest is not a general definition of a particle.

Separately, *T=0* in an evolution record names a prepared state and a clock origin. It is not the short-wavelength endpoint of a spectrum. An ultraviolet cutoff, an initial time and a spatial boundary have different mathematical roles.

A local freely falling observer's null-cone speed, coordinate propagation rate, group velocity, phase velocity and a reduced response delay are also different quantities. A varying coordinate speed near a horizon is not evidence that the local causal constant changes. Claims about different regional *c*'s require an operational, dimensionless comparison and a consistent theory. Ellis and Uzan, gr-qc/0305099v2 explain why the distinct roles often assigned the symbol *c* should be separated. The present atlas uses conventional local null cones; its clock conversion and delayed response do not change that local law.

18. Dark gravitational effects require a stress and an observable

Open physical target. The unexplained gravitational and expansion phenomena motivating the project must be investigated after an exterior model supplies consequences. A prescribed fraction, including “95%,” is not inserted as an exterior abundance or a target that chooses the map.

Clustering and lensing require a stress or effective gravitational response with the right spatial, temporal and scale dependence. Accelerated expansion requires a separate background and perturbation analysis. Antimatter and clock arrows require still other data. The calibration scalar pulse has no calculated energy flux or backreaction and leaves no persistent central field in its certified interval. It supplies none of these dark components.

The foundation's Schur response and the library's shadow-matter and brane examples remain conditional starting points. A successful next connection would carry an actual source/state through the admitted geometry into a gauge-invariant gravitational observable. It would then either distinguish that exterior from alternatives or demonstrate a stated degeneracy. Fitting a freely chosen incident waveform cannot supply that discrimination.

19. Matter, antimatter and orientation

Open physical target. “Inside” and “outside” are geometric labels. Charge conjugation transforms fields and their gauge representations. Changing radial orientation or the sign of a chart coordinate does not, by itself, convert matter into antimatter. A proposed boundary identification must say which field components and charges it matches and whether the source, state and conserved currents permit it.

A real wave's sign change is not charge conjugation. CPT symmetry is also distinct from time reversal alone; neither supplies an entropy arrow.

The library preserves distinct CPT, black-mirror and torsion-production proposals. They cannot be combined by assigning one side of every sphere to antimatter. A useful next test would choose one actual attachment, write its field transformation and determine whether it selects a charge asymmetry from the stated initial data. The current real scalar has no particle–antiparticle charge degree of freedom and cannot settle this.

20. Time, recycling and the next generation

Open physical target. A continuing child needs a domain beyond a single initial slice and a source whose evolution remains meaningful there. The present wall reference has a singular past; the gluing theorem supplies initial data only. Neither is a proof that an astrophysical collapsing black hole forms the mapped cap, or that the cap is stable under disturbances. Regular background curvature and perturbative stability are separate tests.

Proper times can be compared along specified worldlines with a matching convention. Cross-universe ages cannot be read from unrelated coordinate times or unrecorded clock origins. Expansion, entropy increase, causal orientation and an observer's records may all participate in an arrow of time, but a clock conversion alone establishes none of those thermodynamic claims.

Open physical target. Finite-depth inheritance and periodic regional transfer do not prove cosmic recycling or an eternal source. A new generation requires an identified transition, source/state transfer and entropy accounting. Multiple holes motivate multiple possible offspring; they do not require equal children or communication between siblings.

Known branching mathematics should be borrowed. In the finite-type illustration used by Altenberg, 1302.1293v2, an offspring law determines a mean matrix N_{ij}; conditional expected populations evolve by N^n. Its long-time conclusions require the stated branching and spectral hypotheses. The unresolved physics is the transition law and its admissible parameter/state inheritance. Choosing that law to favor an intended cosmic population would assume the result. No infinite-generation computation is needed before one such transition has been established.

Conformal cyclic cosmology remains a separate possible continuation, with its own conformal matching and matter assumptions. It is not identified with black-hole offspring or communicating mouths by a scale analogy.

21. Measurements that could decide between maps

Open physical target. A discriminating prediction must survive changes of units, chart labels and unspecified initial data. The protected vacuum interval proves a finite geometric degeneracy. The scalar inverse in Section 13.3 determines the preparation for a fixed geometry and supplies preparation-class exclusions. It does not establish equivalence between two physical exterior models. Both results identify what information the next observer comparison must supply.

The next admissible calculations therefore have concrete decisions:

Missing connection	Finite decision	What it would unlock
Initial mass response to the wall	Which propagation equations and compatible field/gravitational data carry the sourced geometry to a dynamical wall junction?	A calculated wall response and, subsequently, a corrected observer signal
Preparation-restricted observer inference	Does the source reconstructed from a measured trace satisfy an independently justified preparation law in each candidate?	A discriminating response-range test or an explicit equivalence class
Alternative coupling or source	Does a fixed minimal-coupling or sourced model leave a distinguishable tail?	A comparison of interior response mechanisms
Physical regional Hamiltonian	Does a chosen matched field yield the required reduction and initial covariance?	Memory with an identified geometric exterior
Stability or formation of an attachment	Do fixed-source disturbances or collapse data remain in the claimed domain?	A stronger cosmological interpretation
Dimensionless inheritance	Which transition quantities are preserved or changed?	A meaningful comparison of constants across regions

These are candidates, not a compulsory queue. Each proposed calculation must state what is borrowed, what is missing, what it decides, what it unlocks, its finite domain and budget, and the outcome that stops or redirects it. The existing admission owner keeps that rule. A failed attachment remains visible, while another geometric route can proceed under its own assumptions.

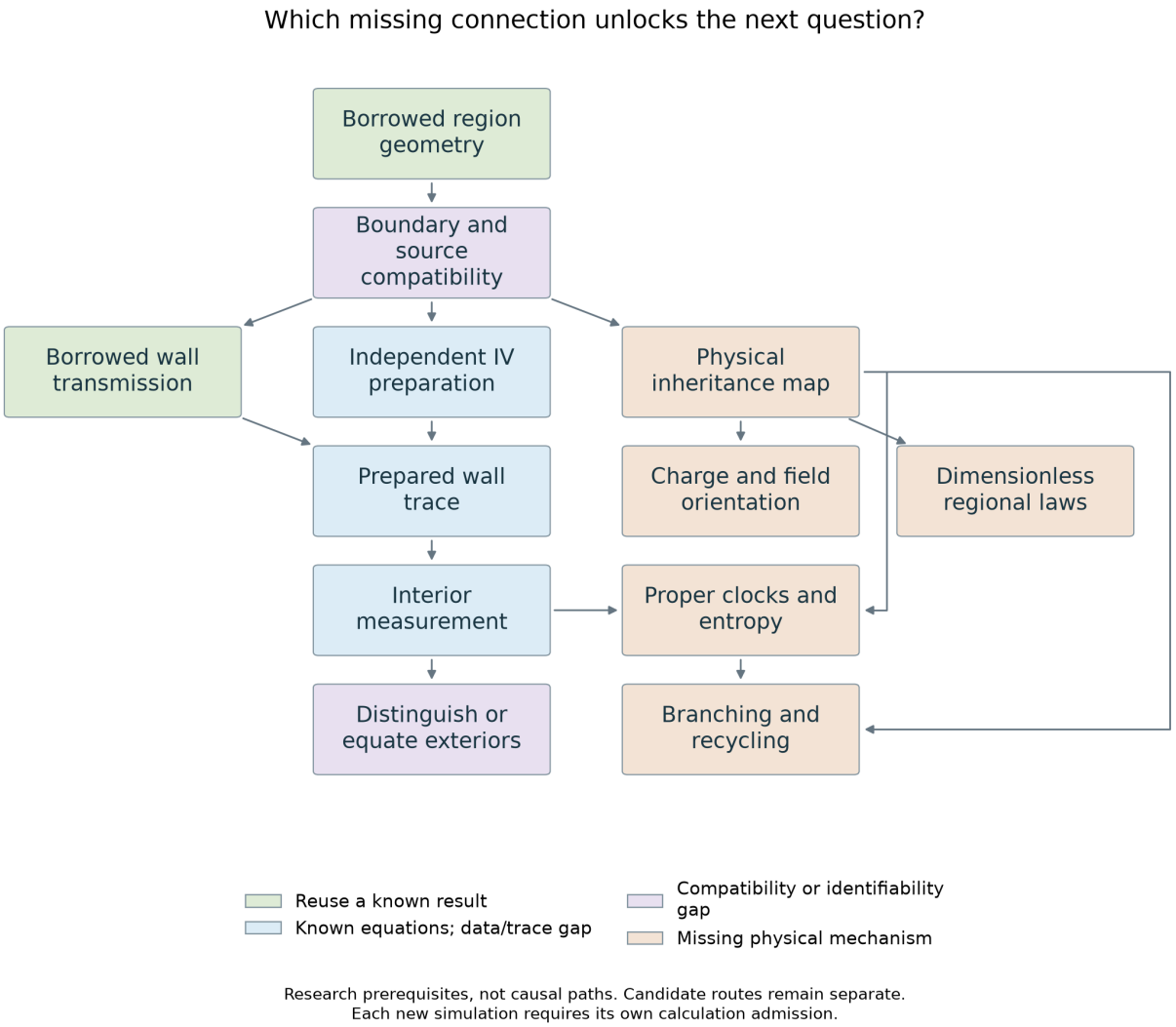


Figure 10. Research dependencies, distinct from spacetime causal arrows. Known equations become usable when their inputs, domains and matching are established. The diagram records the v7 prerequisite structure; the updated programme record names prerequisites, decisions, budgets and stop conditions for eight themes: spherical stability, interior/exterior description, quantum fields, barriers, scales, charge, time and branching. Arrows do not declare these questions solved or prescribe a single compulsory route.

The current result is a continuous mathematical story at a deliberately finite scale: an exterior can be attached on a sourced slice; an isolated completion determines observer access; a specified field state gives an interior waveform; independent finite exterior data now predict a pulse and a positive finite scattering tail; and finite regional reduction explains how unresolved degrees of freedom can retain memory in another fully stated model. The new inverse shows how a complete finite signal selects its preparation within a fixed geometry, and why physical source restrictions must accompany any comparison of maps. The quadratic stress check now connects that preparation to compatible local wall conditions. Its characteristic constraint requires a nonzero initial radius correction under the declared corner and null normalization; transverse constraint transport then supplies an invariant geometric mass coefficient on that same surface. Bulk and wall evolution remain to be established. The remaining work is to connect those statements physically where their domains presently differ.

Appendix A. Evidence, attribution and preserved mathematics

This living manuscript draws from source-bound atlas records without changing their historical bytes. Record residuals and refinement comparisons are reported at their declared scope; floating-point quadrature agreement is not a rigorous error bound. The overview producer reads existing records and renders their relationships without launching a new physics calculation.

Result family	Owning derivation and preserved evidence
Finite inheritance and nested reduction	Proof; snapshot
Two-hole parent and radiation attachment screen	Atlas v2; manifest
Signed moving wall and gluing literature	Atlas v3; manifest
Sourced slice and vacuum observer domain	Atlas v4; manifest
Isolated causal completion and clocks	Atlas v5; manifest
Classical incoming state and scalar transfer	Atlas v6; manifest
Finite exterior preparation and research dependencies	Preparation; programme; v7 manifest
Evaluated exterior preparation and finite tail	Response; v8 manifest
Fixed-model source inverse and preparation exclusions	Proof; record
Conformal probe surface stress and tangential flux	Derivation; record
Characteristic gravitational initialization	Constraint and frame check; record
Geometric mass on the prepared null surface	Mixed geometry and stress checks; record
Finite spherical transfer, feedback and memory	Companion; frozen build/evidence
Foundation boundary, spectral and source constructions	Archived v0.26.0; research inventory

The archived foundation retains, among other strands, its full-spinor boundary maps, scalar-link Dirac benchmark, source and work ledgers, curved spectral constructions and explicitly limited black-universe benchmarks. Their theory, state and domain remain with their owning notes. An RN pressure diagnostic is not promoted to mechanical balance, a reached inner horizon is not promoted to central regularity, and stopped numerical runs without completed summaries supply no new physical conclusion. These distinctions preserve useful work without making historical routes mandatory.

The arXiv library records paper versions, reading depth, reusable equations, assumptions and unresolved connections. It includes the sources for the supplementary multiple-daughter, multi-mouth, CPT, torsion and regular-interior branches. A bounded search cannot certify that no one has already closed an apparent gap. The paper guide locates the unchanged atlas working drafts and the focused companion. The author's responsibility includes the interpretations and synthesis here; borrowed results retain their source attribution and are not presented as new laws.